Suraj Rai

Senior Full-Stack Engineer | Tech Lead

Chitwan, Nepal | +977 984 532 0407 | shuraj.shampang@gmail.com
linkedin.com/in/shuraj-shampang-9ab602b5 | **github.com/shurajcodx**

PROFESSIONAL SUMMARY

Senior Full-Stack Engineer and Tech Lead with 8+ years building and delivering production web, mobile and AI-enabled platforms in healthcare, logistics, digital asset management and platform-as-a-service. Lead development on several products at once: set architecture and design direction, review code, mentor engineers and own delivery with customers, consultants and designers.

Fixed Elasticsearch out-of-memory and slow-search failures by migrating **nearly 1 million records** to a nested mapping with **zero downtime**, cut manual deployment effort by **about 70%** through a CI/CD redesign and reduced end-to-end latency by **about 30%**. Strong in API and service design (Node.js, Spring Boot, Symfony, Laravel), React and TypeScript, DevOps and monitoring, HL7 FHIR healthcare integrations and AI integration. Work across a distributed Nepal and Hong Kong team.

TECHNICAL SKILLS

Back end and APIs	Node.js (Express, Fastify), Spring Boot, Symfony, Laravel, REST, GraphQL, WebSocket, Kafka
Front end	React, TypeScript, Next.js, Vue
Mobile	Native Android, Flutter, React Native
Languages	TypeScript, PHP, Java, Python (scripting)
AI	AI integration, semantic search, MCP server development
Data	PostgreSQL, MySQL, MariaDB, MongoDB, Redis, Elasticsearch
Cloud and DevOps	AWS, Docker, Jenkins CI/CD, Nginx, Grafana, Prometheus
Engineering practice	Architecture and design docs (RFCs, ADRs), code review, Jest, React Testing Library, Cypress, Agile/Scrum, mentoring, HL7 FHIR integration

PROFESSIONAL EXPERIENCE

Dzango Technologies, Hong Kong (Remote)

Mar 2018 to present

Senior Full-Stack Engineer — Nov 2025 to present

- Eliminated recurring Elasticsearch out-of-memory and slow-search failures on the Bimbo platform, caused by the maximum field count, by redesigning the index with a nested approach and migrating **nearly 1 million records with zero downtime**.
- Stabilised servers that crashed from many concurrent file downloads writing temporary files (out-of-memory) by adding caching for asset and file downloads and a download queue, optimising the code and upgrading storage.
- Cut manual deployment effort by **about 70%** by redesigning CI/CD and deployment; built Grafana and Prometheus dashboards for production visibility.
- Reduced end-to-end latency by **about 30%** through render profiling, code splitting, lazy loading, memoisation and API caching.
- Lead development on concurrent products in healthcare, HR and PaaS, setting design direction and reviewing code.
- Own architecture across front end, back end and DevOps, setting folder structure, state-management patterns and conventions that the wider team follows.
- Lead a shared component library and design system used across multiple products, removing duplicated UI code.
- Mentor junior and mid-level engineers through code reviews and 1:1s; write RFCs and technical design docs that shortened onboarding.
- Led AI integration for the Bimbo digital asset platform: LLM pipelines, fine-tuning, semantic search and an MCP server.

PROFESSIONAL EXPERIENCE (CONTINUED)

Full-Stack Engineer II | Dzango Technologies Nov 2022 to Nov 2025

- Acted as feature tech lead, working directly with customers, consultants and designers to turn requirements into accessible interfaces.
- Broke large initiatives into scoped Jira and Notion tickets with clear acceptance criteria.
- Built REST APIs, background jobs and WebSocket integrations for real-time data, and internal reusable component libraries.
- Delivered complex React features using Redux at scale and custom hooks for shared data-fetching and form logic.

Full-Stack Engineer I | Dzango Technologies Jul 2020 to Nov 2022

- Built React and React Native features backed by Node.js and Spring Boot services, with predictable loading, error and edge-case handling.
- Integrated REST and third-party APIs, took part in sprint planning and estimation, and maintained CI/CD pipelines.

Associate Software Engineer | Dzango Technologies Mar 2018 to Jul 2020

- Built UI components and back-end endpoints for enterprise healthcare and logistics applications.
- Wrote unit tests and fixed bugs that stabilised early production releases.

SELECTED PROJECTS

12Deg, Platform-as-a-Service console

Technologies: React, TypeScript, Fastify, PostgreSQL, Docker, WebSocket, AI server

- Built the console for application deployment, environment management and release-workflow automation, streaming real-time deployment status over WebSocket.
- Built a planning layer for briefs, specs and engineering docs (ADRs, RFCs), so planning and delivery live in one platform.
- Built the AI server and fine-tuned models to generate briefs and specs from a project concept.

Bimbo, digital asset management

Technologies: React, Node.js (Express), PHP/Symfony, MariaDB, Elasticsearch, MCP, AI integration

- Designed the full AI pipeline from asset ingestion to model inference to indexed results, covering auto-tagging, metadata generation and semantic search, routing tasks across multiple AI models to balance quality, cost and data control.
- Built a drag-and-drop page builder with a custom CSS Grid layout engine, and an MCP server that lets users create and publish pages through Claude (beta).
- Implemented Elasticsearch search and faceted filtering, with workflows for permissions, bulk actions and large result sets.

Embleema, healthcare and clinical research platform

Technologies: React, React Native, Redux, Express.js, Spring Boot, MariaDB, MongoDB, Redis, HL7 FHIR

- Built interfaces for secure patient data collection and decentralised clinical workflows, with multi-step forms, validation and role-based views over HL7 FHIR data models.
- Used Redis caching to speed up data-heavy screens.

Airwaybill, logistics platform

Technologies: React, Spring Boot, MariaDB, Android (Java), iOS (Swift)

- Built shipment lifecycle, tracking and operational workflow modules, and dashboards backed by REST APIs for real-time logistics processing.

EDUCATION

BSc in Information Technology, Sikkim Manipal University, Gangtok, India Apr 2014 to Apr 2018